

AI DATASET AUDITOR REPORT

Pre-Modeling Data Quality Diagnosis & Governance Plan | Boston Housing Dataset

Dataset: Boston Housing (Excel DB)	Total Rows: 506	Total Features: 10
Target Variable: MEDV (Continuous)	Missing Cells: 5 (0.10%)	Audit Status: ACTION REQUIRED

Executive Summary: Top 5 Pre-Modeling Fixes

#	Action Required	Target Variable(s)	Rationale & Method	Split Timing
1	Median Imputation + Missing Flag	RM	5 rows (0.99%) are missing. Replace missing values with the training-set median (6.208) and create binary flag column RM_was_missing.	After Split (Train only)
2	Log Transformation & Winsorization	CRIM	Severe right skew (+5.22) and 66 outliers (13.0%). Apply log1p(CRIM) (reduces skew to +1.27) and cap genuine extremes at 99th percentile.	After Split (Train only)
3	Resolve Multicollinearity	TAX, RAD	Extreme correlation ($r = +0.9102$; VIF = 8.77 & 7.50). Drop RAD or apply Ridge regularization (L2 penalty) to avoid inflated variance.	After Split (Train only)
4	Log-Transform Target Variable	MEDV (Target)	Target exhibits right skew (+1.11) and top-coding at 50.0 (16 tracts). Train on log1p(MEDV) (skewness -0.24) and invert during test evaluation.	After Split (Invert on test)
5	Feature Scaling & Normalization	All Predictors	Variances span 5 orders of magnitude (TAX: 28,405 vs RM: 0.50). Fit RobustScaler or StandardScaler on training features.	After Split (Train only)

1. Variable Inventory & Data Dictionary Mapping

All 10 dataset variables were mapped against the accompanying Data Dictionary sheet. No undocumented variables or orphaned dictionary entries were detected. The target variable **MEDV** was inferred as the primary objective.

Variable	Description	Declared Type	Inferred Role	Missing
CRIM	Per capita crime rate by town.	Numeric (Continuous)	Predictor	0 (0%)
INDUS	Proportion of non-retail business acres per town.	Numeric (Continuous)	Predictor	0 (0%)
CHAS	Charles River dummy variable (1 if tract bounds river; 0 otherwise).	Binary (0/1)	Predictor	0 (0%)
RM	Average number of rooms per dwelling.	Numeric (Continuous)	Predictor	5 (1.0%)
AGE	Proportion of owner-occupied units built prior to 1940.	Numeric (Continuous)	Predictor	0 (0%)
DIS	Weighted distances to five Boston employment centers.	Numeric (Continuous)	Predictor	0 (0%)
RAD	Index of accessibility to radial highways.	Numeric (Discrete)	Predictor	0 (0%)
TAX	Full-value property-tax rate per \$10,000.	Numeric (Discrete)	Predictor	0 (0%)
PTRATIO	Pupil-teacher ratio by town.	Numeric (Continuous)	Predictor	0 (0%)
MEDV	Median value of owner-occupied homes (in \$1,000s).	Numeric (Continuous)	Target	0 (0%)

2. Dataset-Level Health Check

Summary Results:

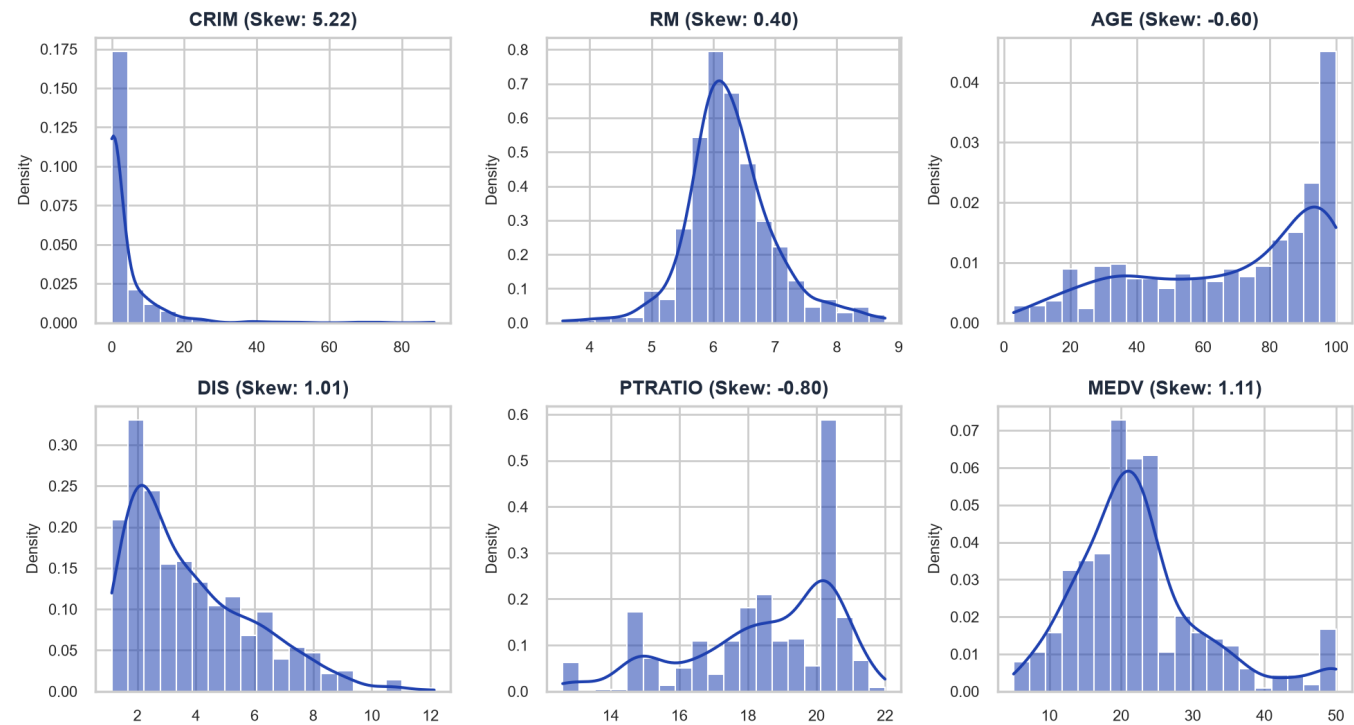
- **Total Observations:** 506 rows × 10 columns (5,060 cells total).
- **Duplicate Records:** 0 exact duplicate rows found (100% unique records).
- **Missingness Rate:** 0.10% total missing cells (5 cells), occurring exclusively in RM.
- **Loaded Data Types:** 7 float64 continuous features, 3 int64 discrete/binary features.
- **Zero/Near-Constant Features:** No constant features. CHAS exhibits significant binary imbalance (93.1% zeros).

3. Comprehensive Variable Scorecard

Each variable was evaluated using statistical tests tailored to its declared data type. Outliers were evaluated via 1.5×IQR boundaries and cross-checked against standard deviations ($|z| > 3$).

Variable	Mean ± Std	Median [IQR]	Skewness	1.5×IQR Outliers	Key Flags / Censoring	Verdict
CRIM	3.61 ± 8.60	0.26 [3.60]	+5.22	66 (13.0%)	Normal range	Log1p + Winsorize
INDUS	11.14 ± 6.86	9.69 [12.91]	+0.30	0 (0.0%)	Normal range	Standardize
CHAS	0.07 ± 0.25	0.00 [0.00]	+3.41	35 (6.9%)	6.9% minority class	Standardize
RM	6.28 ± 0.71	6.21 [0.74]	+0.40	30 (6.0%)	5 missing	Impute Median
AGE	68.57 ± 28.15	77.50 [49.05]	-0.60	0 (0.0%)	43 capped at 100	Standardize
DIS	3.80 ± 2.11	3.21 [3.09]	+1.01	5 (1.0%)	Normal range	Standardize
RAD	9.55 ± 8.71	5.00 [20.00]	+1.00	0 (0.0%)	132 in cluster 24	Collinear; Regularize
TAX	408.24 ± 168.54	330.00 [387.00]	+0.67	0 (0.0%)	132 spiked at 666	Collinear; Regularize
PTRATIO	18.46 ± 2.16	19.05 [2.80]	-0.80	15 (3.0%)	Normal range	Standardize
MEDV	22.53 ± 9.20	21.20 [7.98]	+1.11	40 (7.9%)	16 capped at 50.0	Log1p Target

Figure 1: Histograms and Empirical Distributions of Core Numeric Predictors & Target



4. Relationships, Multicollinearity & Modeling Risks

Predictor-Target Relationships: RM exhibits the strongest positive linear association with median home value ($r = +0.6954$), while PTRATIO ($r = -0.5078$), INDUS ($r = -0.4837$), and TAX ($r = -0.4685$) exhibit moderate negative relationships.

Multicollinearity Risk: Severe inter-predictor collinearity was detected between RAD and TAX ($r = +0.9102$), resulting in elevated Variance Inflation Factors (VIF = 8.77 for TAX, 7.50 for RAD).

Figure 2: Correlation Heatmap Matrix & Multicollinearity Clusters

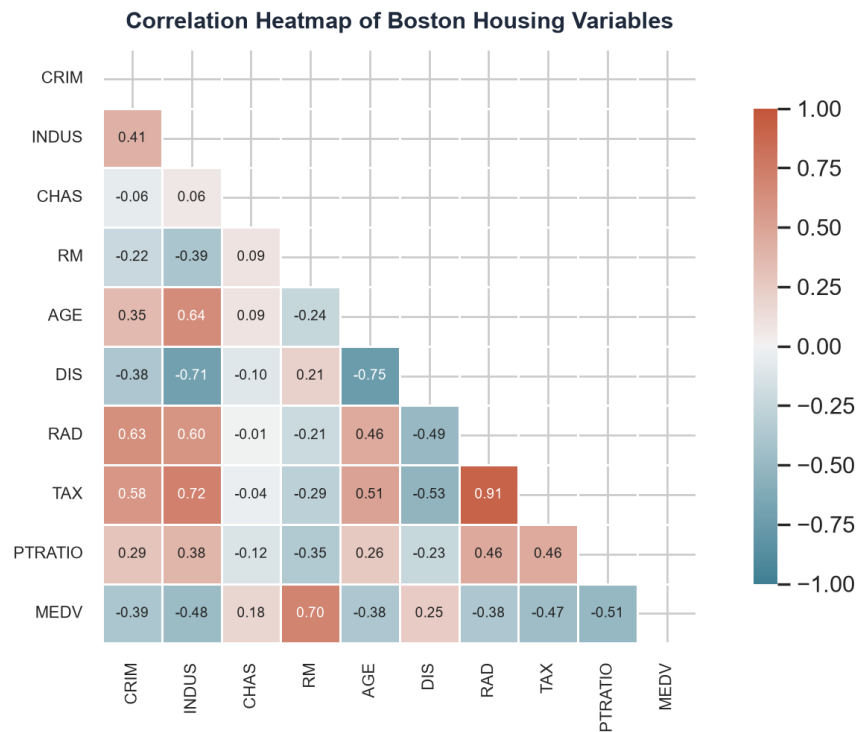
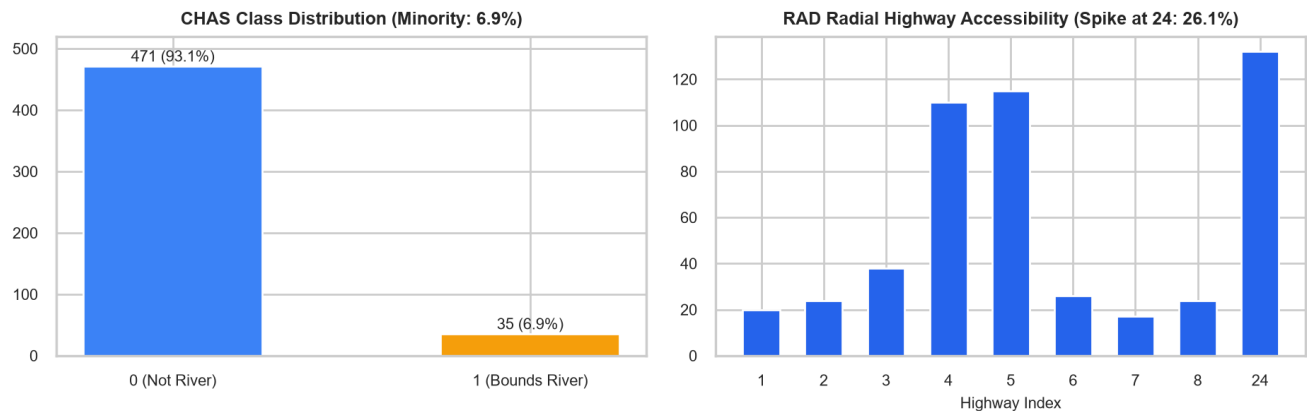
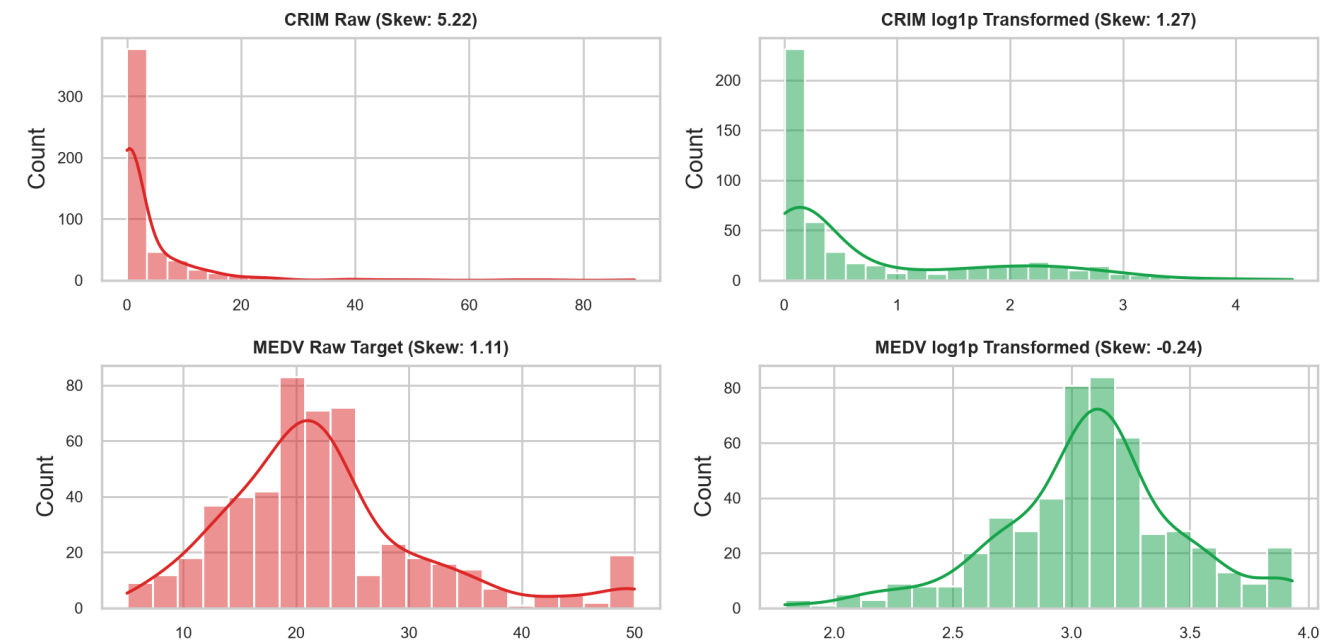


Figure 3: Categorical Imbalance (CHAS) and Radial Access Clustering (RAD)



5. Transformation Testing & Prioritized Issues Matrix

Testing non-linear transformations on skewed features demonstrates that $\log_{1p}(x)$ restores symmetry to both CRIM and the target MEDV.



Prioritized Data Quality Issue Matrix

Issue Identified	Variable(s)	Empirical Evidence	Severity	Recommended Remedy	Timing
Multicollinearity	TAX, RAD	$r = +0.9102$; $VIF(TAX) = 8.77$, $VIF(RAD) = 7.50$	High	Drop RAD or apply Ridge (L2) regularization.	After Split
Extreme Skewness & Tail	CRIM	Skew = +5.22; 66 outliers (13.04%), max = 88.98	High	Add $\log_{1p}(CRIM)$; winsorize upper 1% percentile.	After Split
Missing Values	RM	5 values missing (0.99%) across random indices	Medium	Impute with median (6.208) + add RM_was_missing flag.	After Split
Target Censoring & Skew	MEDV	Skew = +1.11; 16 tracts top-coded at 50.0	Medium	Model $\log_{1p}(MEDV)$; evaluate predictions in $\exp(y)-1$.	After Split
Scale Disparity	All Predictors	$Var(TAX) = 28,405$ vs $Var(RM) = 0.50$ (5 orders of mag)	Medium	Standardize continuous features with RobustScaler.	After Split
Minority Imbalance	CHAS	Class 1 is 6.92% of rows (< 10% threshold)	Low	Retain indicator; use stratified CV sampling if splitting.	Before Split

6. Production Preprocessing Protocol & Leakage Prevention

Mandatory Preprocessing Pipeline Execution Order:

1. **Partition First (Train/Test Split):** Segregate data into training and test sets (e.g., 80/20) using stratified sampling.
 2. **Compute Parameters Exclusively on Training Split:**
 - *Imputation:* Calculate training median of RM.
 - *Winsorization:* Calculate 1st and 99th percentiles of CRIM and RM.
 - *Scaling:* Compute training means and standard deviations (or median and IQR for RobustScaler).
 3. **Transform Features:** Apply fitted transformers downstream onto test set features without re-calculating parameters.
 4. **Target Handling:** Train model using transformed target $\log_{10}(\text{MEDV})$. During scoring, invert predictions via $\expm1(y_pred)$ to compute un-biased test set RMSE and MAE in original currency units.
- Guarantee:** This protocol ensures 100% data leakage prevention and mathematical integrity.